



K Madhunisha

cb.en.u4cce23020@cb.students.amrita.edu | 9629275186 | 2027

B.Tech, Computer and Communication Engineering

Address : Potthan Thottam, Rangappanaickenpalayam, Namakkal

Education

ASE, Coimbatore

B.Tech · Computer and Communication Engineering

2027

CGPA - 7.41/10

NATIONAL PUBLIC SCHOOL

Class XII - CBSE · mbipc · VEEPANATHAM NAMAKKAL TN (55187)

2023

Percentage - 76.4%

Dr.M.S. Udayamurthy School of Excellence

Class X - CBSE · mbipc · Namakkal

2020

Percentage - 75%

Experience

Emertxe Information Technologies | [Link](#)

IoT Intern · Internship

Sep 2025 - Nov 2025

Online

- Developed IoT-based smart applications including a Home Automation System using Blynk IoT.
- Utilized PICSimLab for simulation, testing, and validation of microcontroller-based projects.
- Applied Embedded C programming for sensor integration, device communication, and automation tasks.
- Gained hands-on experience in IoT architecture, cloud connectivity, debugging, and SDLC-based project development.

Projects

AI Book Analyzer | [Link](#)

Self-Initiated Project · PyPDF2,NLTK,Pandas,NumPy,Plotly,NetworkX

Jan 2025 - Feb 2025

Artificial Intelligence, NLP

- Developed an AI-powered application to analyze PDF books and generate concise summaries, FAQs, and interactive question-answer responses using **Natural Language Processing (NLP)**.
- Implemented PDF text extraction and preprocessing to accurately process documents and improve the quality of AI-generated responses.
- Integrated **Large Language Models (LLMs)** to provide context-aware summaries, answer user queries, and generate meaningful insights from uploaded books.
- Designed an intuitive web interface that allows users to upload PDF documents and interact with the content through a chatbot-like experience.
- Technologies Used: Python, Streamlit, Google Gemini API, PyPDF2, NLP, LLMs.

Supply Chain Management System

Academic Project · DBMS,SQL,System Design

Dec 2024 - Apr 2025

Database Management System

- Developed a centralized **Supply Chain Management System** to manage inventory, supplier records, purchase orders, and logistics information.
- Designed normalized database schemas and implemented **SQL** operations for data storage, retrieval, reporting, and transaction management.
- Improved inventory visibility and order tracking by integrating supplier and logistics data into a single platform.
- Technologies: SQL, DBMS, Database Design, System Design.

Smart Trolley with Automated Billing System

Academic Project · SPI Communication ,Embedded C/C++

Jun 2025 - Mar 2026

Internet of Things

- Developed the application logic using **Arduino C/C++** to manage product scanning, cart operations, bill calculation, and checkout functionality.
- Implemented **RFID-based product identification** and integrated load cell validation to ensure accurate product detection and prevent billing errors.
- Designed the software for **real-time communication** between the Arduino and ESP32, enabling bill updates and digital receipt delivery over Wi-Fi.
- Performed debugging, testing, and code optimization to improve system reliability, handle repeated product scans, and ensure smooth integration of all software modules.
- Technologies Used: Arduino C/C++, Arduino IDE, ESP32, Embedded Programming, RFID (MFRC522), HX711 Load Cell.

Skills

Areas of Interest: Natural Language Processing , AI & Communication System , Data Science & Analytics

Languages and Frameworks: Numpy, PyPDF2,Pandas, React

Programming Languages: Python, SQL, HTML, CSS

Tools and Technologies: Visual Studio Code, GitHub , Jupyter Notebook, MySQL, Arduino

Languages

English [Professional Working Proficiency], Tamil [Native Proficiency]